

# Sentiment-Driven Gold Price Prediction

An NLP Approach to Financial News and Gold Trading Signals

**Team Members:**

Arko Bhattacharya

Zachary Fennie

Vishesh Gupta

Danish Maknojia

Chris Moreira

**Team Identifier:** 3 - Chameleon

# Abstract

This project presents a modeling approach for gold price moves that integrates traditional market data with unstructured news headline data. We first establish baseline models - a Temporal Convolutional Network (TCN) and a Long Short-Term Memory (LSTM) - using only price-related features. We progressively enhance these models by incorporating signals extracted from 10,000 gold-related news headlines. We extract our sentiment scores using FinBERT and latent topic embeddings and generate clusters using unsupervised machine learning, which capture market mood and prevailing narratives. Since our selected model applied on price data often introduces over-smoothing, traditional metrics like mean absolute error (MAE) can be misleading [1]. This is why we shift our focus to directional accuracy as it would better capture the models effectiveness. Our results show that augmenting the baseline with sentiment and thematic news features yields better forecasting accuracy and model robustness. Incorporating qualitative news information improves the model's ability to predict gold price moves, highlighting the potential of Natural Language Processing (NLP) in enhancing financial prediction models.

# Introduction

Gold, the world's oldest financial asset [5], remains critical with wide-reaching implications across finance, investment, and policy making. As a global benchmark for economic stability and a safe haven during market volatility, gold maintains unique value due to its immunity to inflation and monetary policy fluctuations [6]. Financial forecasting is further complicated by the Efficient Market Hypothesis (EMH), which posits that asset prices fully reflect all available information. Therefore, we believe by incorporating sentiment-related features, we aim to enhance directional accuracy, which, if successful, could potentially be leveraged in trading strategies to capture market inefficiencies. While technological advancements in machine learning and artificial intelligence have enhanced prediction capabilities by processing vast datasets and identifying non-linear relationships, forecasting remains challenging due to gold's sensitivity to unpredictable geopolitical events and global economic forces [7].

Our research takes a fourfold approach that contributes to the expansion of alternative data use in financial forecasting by constructing a comprehensive framework that combines NLP techniques with traditional market analysis, bridging the gap between qualitative market sentiment and quantitative price prediction models.

First, we build a baseline model using historical features to predict the percentage change of gold prices. Next, we improve the model with sentiment scores derived from gold related headlines using NLP techniques. To ensure we don't analyze sentiment in isolation we go further to identify the overarching themes that may reflect the broader market narratives thereby adding another layer of context. At each step above we compare the updated model to our original baseline to assess the improvements in predictive accuracy and trend classification. Finally we combine all our insights into a comprehensive predictive model aiming to capture both quantitative patterns and qualitative market signals for a more robust gold price forecasting. We believe that in this increasingly data rich and sentiment driven market landscape, being able to understand public perceptions' impact on prices is more critical than ever.

## Background

As a consequence of its global importance, gold price forecasting is not a novel idea; however, the incorporation of media sentiment is still nascent. While researching the field, we found two potential models worth exploring: the Long Short-Term Memory (LSTM) network and the Temporal Convolutional Network (TCN). First explored as a means to capture temporal dependencies by enforcing constant error flow [2], the LSTM overcomes the major limitations of static and autoregressive models, which struggle with vanishing gradients, short-term memory, and poor adaptability to complex, evolving sentiment signals [2,3]. Furthermore, the TCN is a suitable replacement for recurrent models to predict gold futures prices from sentiment analysis. Studies have shown that TCNs can be more accurate and efficient than traditional recurrent networks for financial time-series forecasting [8]. The same study also found that the TCN is comparable and competitive with the LSTM, even at times outperforming it [8]. For these reasons, the LSTM and TCN were a natural choice for comparison to determine our baseline model.

Given the limitations imposed by EMH, we explored sentiment-based features as a way to capture market expectations. Abdou et al. [9] explored gold price prediction using sentiment analysis of social media, showcasing a potential, albeit a weak one, of sentiment signals to improve traditional time-series models. Bouri et al. (2015)[10] delved deeper into the relationship between investor sentiment and gold returns, providing evidence that market mood can influence gold price movements. However, a clear signal of sentiment's effect on price movement remained elusive until Hajek's findings that using financial news specifically, through the use of the NLP model FinBERT, performed well [11]. Consequently, we chose to conduct our sentiment-based experiments using FinBert, whose pretraining on financial corpora allows it to capture nuanced sentiment in financial news and reports more effectively than other sentiment models [4]. However, during our analysis, we noticed that these headlines have overarching themes. To tackle this, our approach to thematic clustering in financial headlines focused on emphasising the value of textual analysis for market prediction, which is grounded by prior research. Grootendorst [12] introduced BERTopic, which inspired us to use transformer based embeddings combined with dimensionality reduction and density-based clustering to discover nuanced themes in financial news. Nassirtoussi et al. (2014) [13] and Schumaker & Chen (2009) [14] further influenced that we must pay attention to the role of textual signals in financial forecasting and the need for scalable systems to process unstructured news. Lastly, studies such as Henry (2008) [15] and Goyal et al. (2019) [16] highlighted how surface-level biases and linguistic framing can distort analysis. Together these studies informed our design of a pipeline for an unsupervised thematic discovery, macro-group representation of financial news.

## Data

We used a gold-focused financial news dataset introduced by Sinha and Khandait (2020, 2021) [17], sourced via Kaggle [18]. This is a news dataset for the commodity market where they have manually annotated 10,000+ news headlines across multiple dimensions into various classes. The dataset has been sampled from a period of 20+ years (i.e. 2000-2021). We did our analysis by visualizing the gold open price over time tracing its movements across two decades obtained through Yahoo! Finance (yfinance). During our exploratory data analysis, we found that the percentage change to be

approximately 0% with an equal, normal distribution, illustrated by our Figure 1 below. The result would become impactful during our baseline model discovery as the models’ accuracy also hovered around the same range.

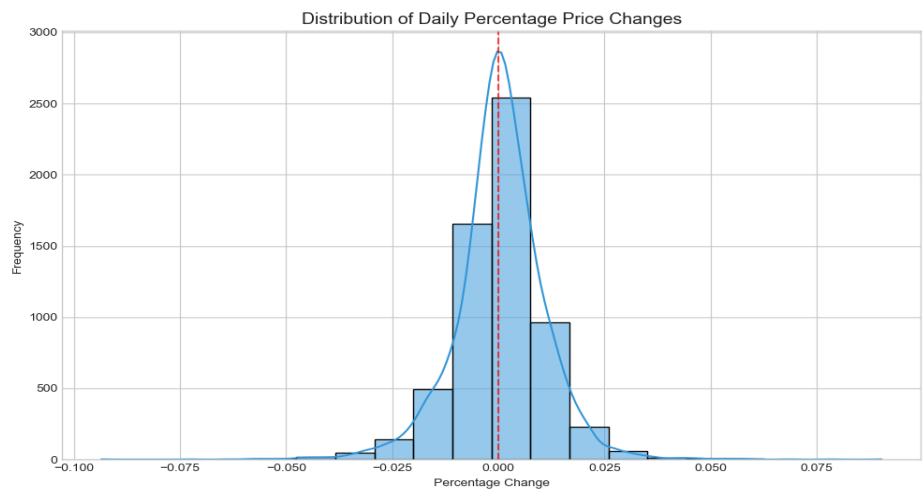


Figure 1: Distribution of Price Percent Change in Yahoo! Finance Dataset

To connect the news data with the market we overlaid red dots onto the Figure 2 below to indicate days when news headlines were recorded. This helped us see the distribution of news relative to significant market activity.

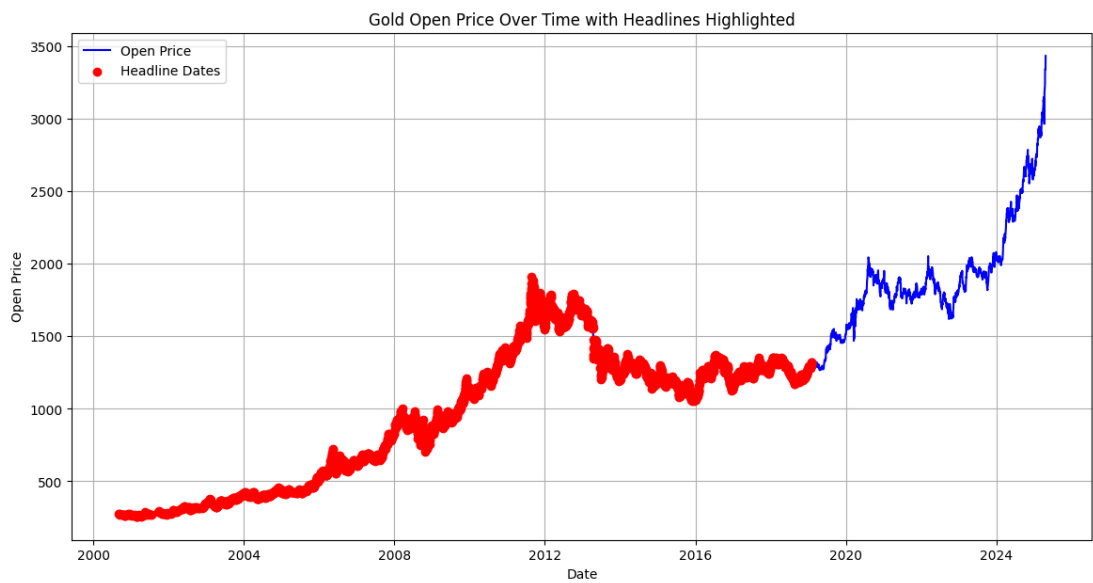


Figure 2: Gold Opening Price 2001 - 2025

We then aggregated and plotted the number of headlines per month which showed periods of heightened news coverage as seen in Figure 3 below.

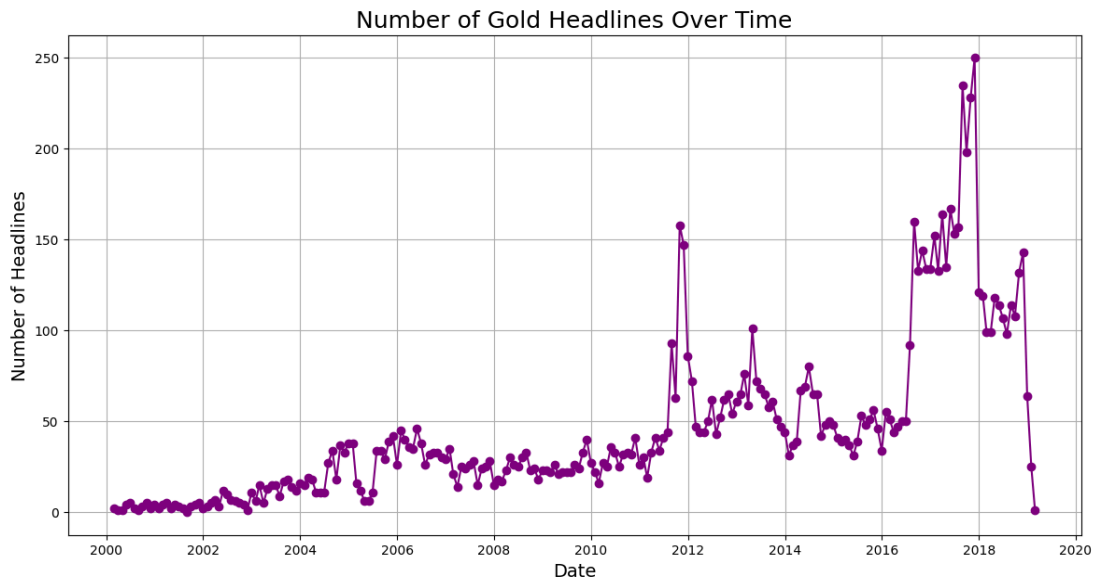


Figure 3: Change of Gold Headlines 2000 - 2019

Lastly, we generated a word cloud across all collected headlines to uncover dominant themes and frequent terms, offering an initial glimpse into the types of narratives that surrounded gold prices as seen in Figure 4.

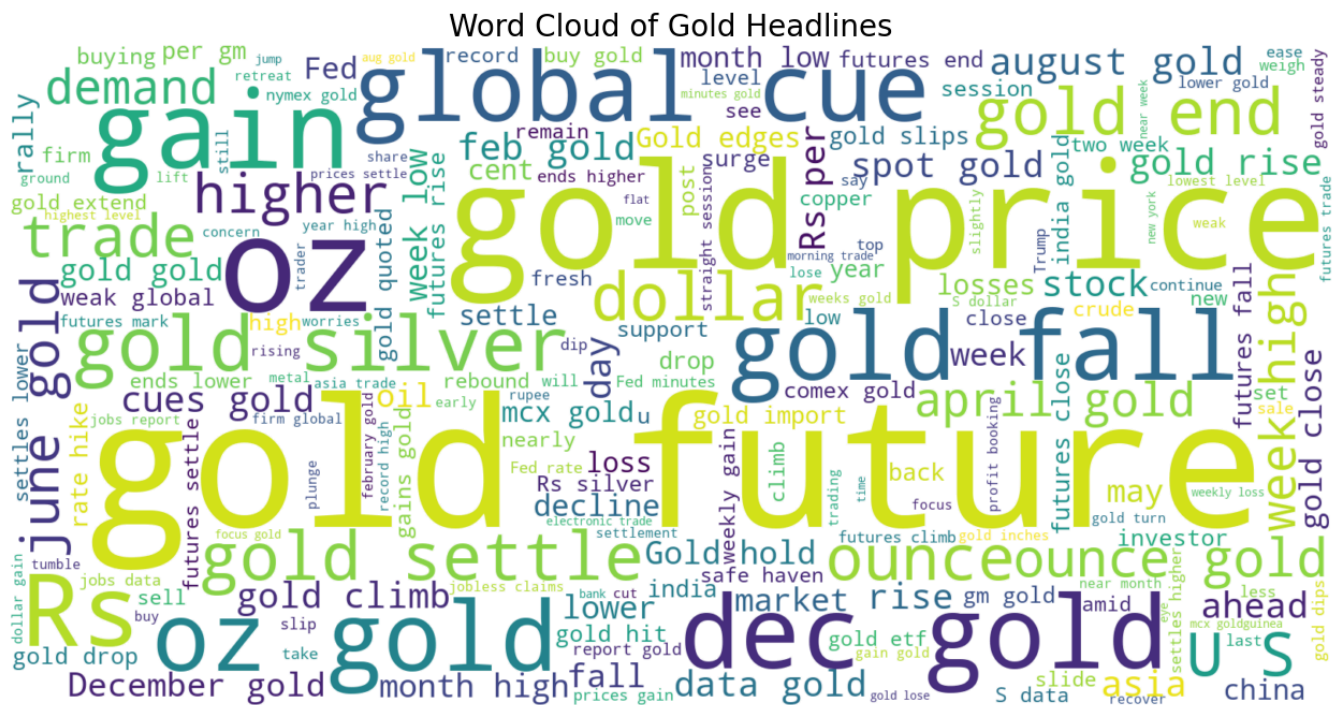


Figure 4: Gold Headlines in Financial News Word Cloud

These analyses helped us understand our news data better, laying an important foundation for deeper modeling and interpretation. Throughout our experiment you will see how we use our data sources above to extract features like sentiment scores, thematic clusters, and price change.

For real-time prediction, we used the Newsdata.io API to fetch three days worth of global news headlines containing the keyword "gold". This retrieved approximately 589 articles which were then cleaned and filtered to retain only those that pertain specifically to gold as a physical commodity or its role within the broader economic context. These inputs are processed through the same feature extraction and preprocessing pipelines used during training, allowing our models to generate next-day gold price movement predictions.

## Experiments

Our project traces the evolution from relying solely on historical gold prices to leveraging the rich, unstructured world of financial news as a predictive signal. We begin by constructing a price-only baseline model to assess whether past gold prices can effectively predict future percentage changes. We then augment this with sentiment scores extracted from over 10,000 financial news headlines to evaluate their added predictive value. Upon closer inspection of the headlines, we noticed certain contextual patterns. To understand these patterns, we turned to unsupervised clustering with BERTopic to reveal latent semantic themes, organizing them into both dominant and macro-level groups.

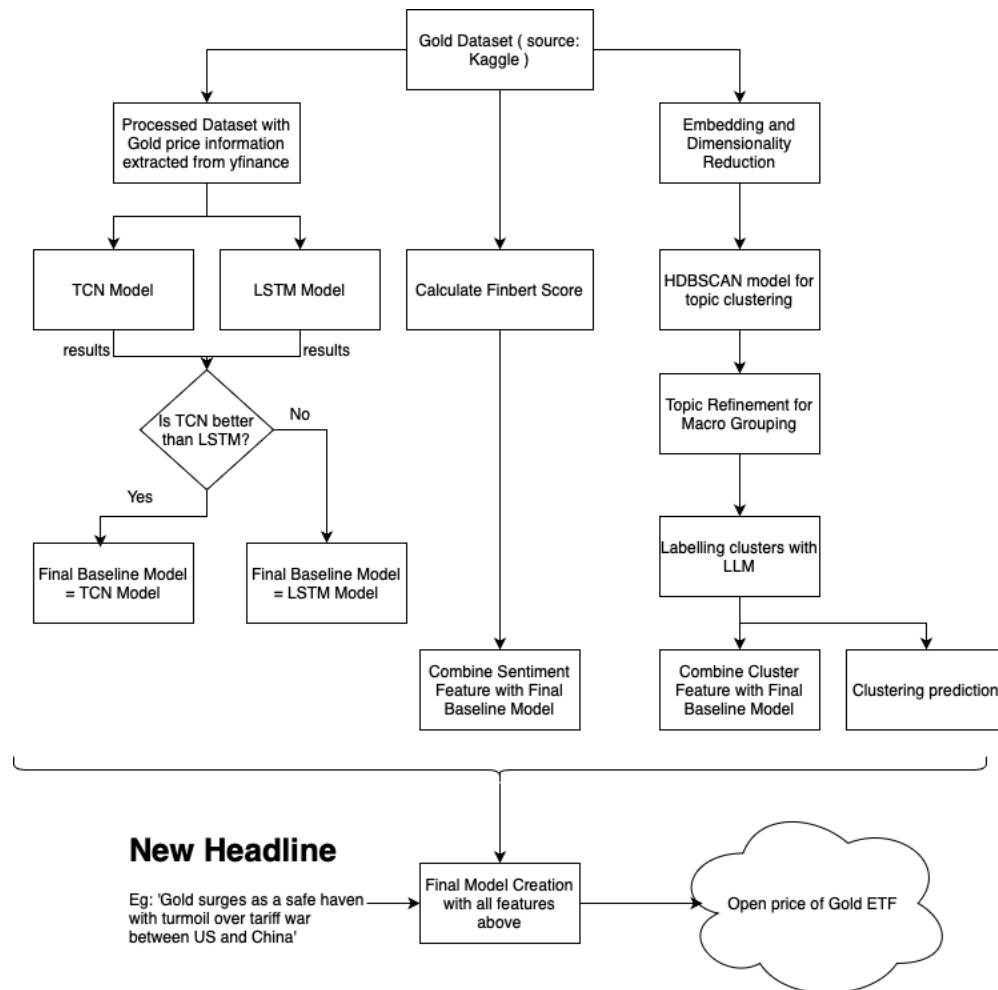


Figure 5: Workflow Overview: From Historical Prices to Thematic News-Based Prediction

We then work on combining all the explanatory variables used in each of the independent paths depicted in Figure 5 into a unified feature set that serves as the input for our final predictive model. These thematic features, capturing the nuanced and multi-layered nature of financial reporting, were then integrated alongside sentiment and price to enhance forecasting accuracy within a single, cohesive predictive framework.

## Experiment 1:

### Baseline Gold Price Prediction Model Discovery

Given the sequential nature of time series data and the complexity in predicting financial price movements, we began with a Long Short-Term Memory (LSTM) network [19]. Using a 3-day lookback window, we evaluated the daily gold price data from 2000 to 2019. While the model captured broad directional trends, it tended to underestimate the full volatility of market movements (Appendix 1). To compare our above approach, we then used a Temporal Convolutional Network (TCN) [21], which we understood has gained traction in time series forecasting due to its interpretability and efficiency. Although both models showed limitations as seen in the Table 1.1 below, the LSTM exhibited marginally better performance across key metrics and maintained more stable training behavior (Appendix 2). We ran our models 20 times, and for 13 of the model runs, the LSTM continued to be marginally better. These findings guided our decision to select the LSTM as our baseline for further development.

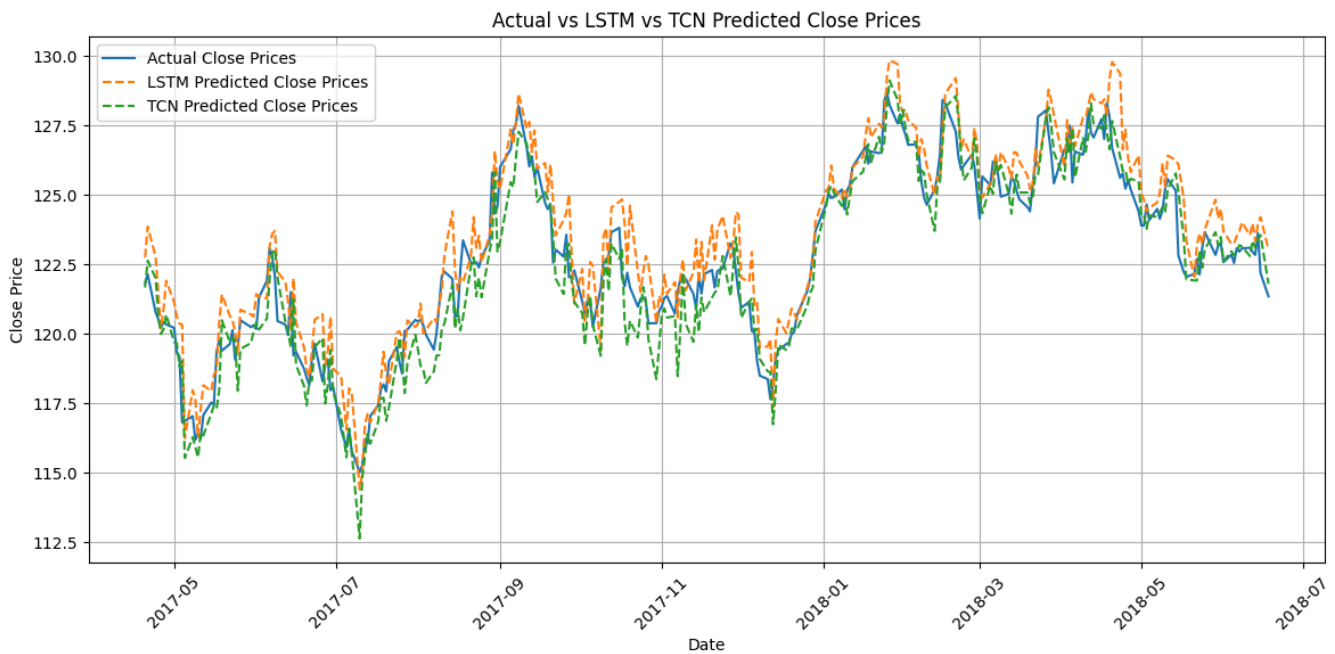


Figure 6: LSTM and TCN Performance: Actual Percent Change vs. Predicted Percent Change

Table 1.1. TCN vs LSTM Performance Metrics

| Model | MAE    | MAE % | Directional Accuracy (%) |
|-------|--------|-------|--------------------------|
| TCN   | 1.9764 | 1.03% | 47.10%                   |
| LSTM  | 1.5013 | 0.78% | 47.70%                   |

As seen in Table 1.1 and Figure 6, both models exhibited limitations (Appendix 14), and while LSTM showed marginally better average performance across key metrics and more stable training behavior (Appendix 2), the performance differences are within potential variability bands commonly observed in deep learning models [25]. These findings guided our decision to select the LSTM as our baseline for further development.

Building upon the insights we gained from our baseline model comparison, we selected the LSTM architecture as our foundation for creating the final predictive model. To enhance the relevance and applicability of our model, we shifted our focus from gold futures to the Gold Shares ETF with the ticker symbol "GLD," which tracks the market price of gold and eliminates the need to manage contract rolls, simplifying the analysis and reducing potential sources of noise.

Unlike static datasets, time series data exhibit dependencies across overlapping periods. By applying a windowing transformation to our dataset and segmenting the 'open' price column into overlapping windows we enable the model to learn from sequences rather than individual data points. This is essential for capturing momentum, trend shifts or even lagged effects (Appendix 5). Additionally, applying a sliding window-based mechanism ensures each window is independently processed, preventing data leakage, decomposition-level inconsistencies, and overly optimistic forecasting performance[20]. Once we have the data windowed, we extract and organize three key components i.e. dates, input sequences, and corresponding targets. We reshape the feature matrix to a 3D array making it suitable for training our LSTM model (Appendix 6).

This allows us to bridge the gap between the featurizing engineering we did before and the modeling training we will perform shortly. The data is then split, preserving chronological order, ensuring that future data is never used to inform past predictions (Appendix 7). After our initial analysis and required set up we go ahead and build our desired LSTM network to predict our future gold prices (Appendix 8). After running our model on the three datasets, we observed that it performed well across all, achieving high accuracy and low error metrics, as shown in Table 1.2 below.

Table 1.2. Updated LSTM Performance Metrics

| Model          | MAE    | MAE % | Directional Accuracy (%) |
|----------------|--------|-------|--------------------------|
| LSTM (revised) | 0.5984 | 0.49% | 47.77%                   |

While there was a slight decrease in performance on the validation and test sets, the model demonstrated strong generalization, effectively handling unseen data (Figure 7).

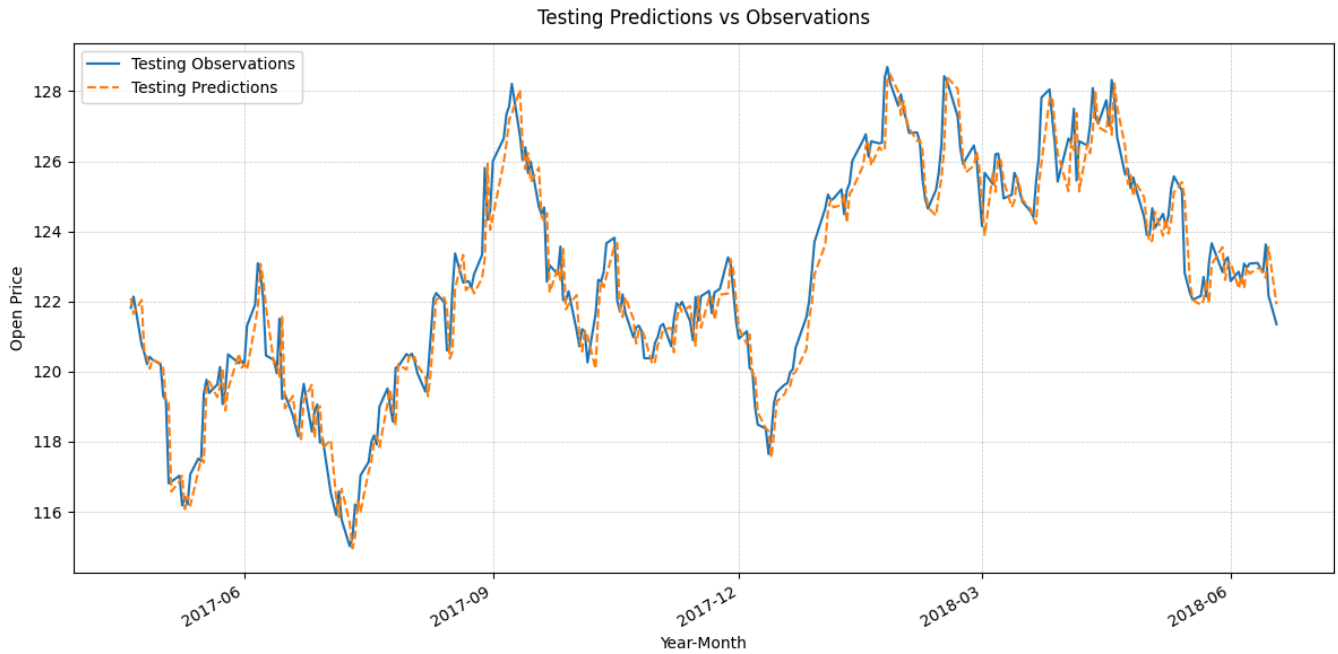


Figure 7: Updated LSTM Performance: Validation vs Test set

## Experiment 2:

### Semantic Analysis and Application on Baseline model

Next, we aim to track sentiment relevant for financial analysis and potential correlation with market movements. We work on obtaining the daily sentiment analysis on financial news using a FinBERT model. We process our dataset by grouping news headlines by date, apply FinBERT to each sentence and average the logits (appendix 16) for each day to derive a sentiment score. The resulting probabilities for positive, neutral, and negative sentiments are calculated using SoftMax, and the sentiment with the highest probability is assigned as the final daily sentiment. We then incorporate this sentiment data into our baseline model described above, which uses open price as a predictor, and compare model performance to evaluate the added predictive value of sentiment.

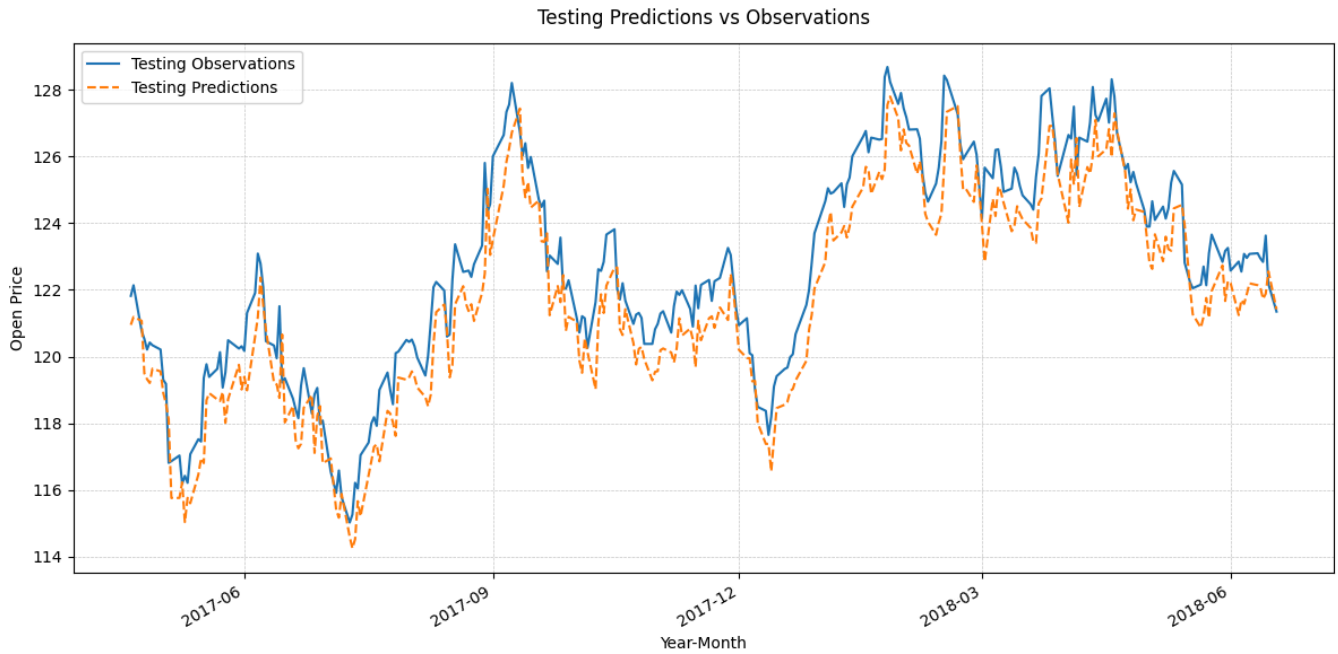


Figure 8: LSTM with Sentiment Performance: Validation vs Test set

When we evaluated both on the testing period the enhanced model demonstrated improved direction accuracy as seen in Table 2.1. This suggests that the updated model was better at predicting the trend of the target variable, potentially indicating that sentiment information contributes positively to identifying market direction.

Table 2.1. Baseline Model vs Baseline with Sentiment Performance Metrics

| Model       | MAE    | MAE % | Directional Accuracy (%) |
|-------------|--------|-------|--------------------------|
| Baseline    | 0.5984 | 0.49% | 47.77%                   |
| w/Sentiment | 1.0833 | 0.88% | 50.86%                   |

Recent literature supports this view, showing that investor sentiment significantly influences asset prices beyond fundamentals [26, 27] and that incorporating news-based sentiment improves forecasting accuracy compared to models relying solely on historical price data [28, 29].

### Experiment 3:

#### Thematic Clustering of Gold-Related Financial News Headlines for Price Signal Extraction along with Application on Baseline model

Building on our prior work that focuses on individual headline sentiment we now explore if thematic cluster can offer a more holistic and structured understanding of various macroeconomic influencing gold markets. This shift allows us to capture patterns that are not immediately apparent in isolated

headlines but may emerge when similar topics evolve over time. By looking at data through a thematic lens we could support our traditional financial indicators with an additional data driven insight into market sentiment.

To minimize surface-level bias, we preprocess headlines to strip reactive language, allowing downstream models to focus on underlying economic signals (Appendix 9). For theme extraction, we use the BERTopic framework [12], which integrates transformer-based embeddings with topic modeling. Unlike traditional methods like TF-IDF or LDA that lack contextual awareness, BERTopic employs the *all-mpnet-base-v2* sentence transformer to capture nuanced financial language and semantic relationships. To enhance clustering quality, we reduce embedding dimensionality using UMAP and apply HDBSCAN, a density-based algorithm effective in detecting variable-sized clusters in noisy data. BERTopic also offers soft probability assignments over topics, enabling us to account for the multi-thematic nature of financial headlines (Appendix 10). The dendrogram in Figure 9 visualizes hierarchical topic relationships, where lower merge distances indicate greater semantic similarity.

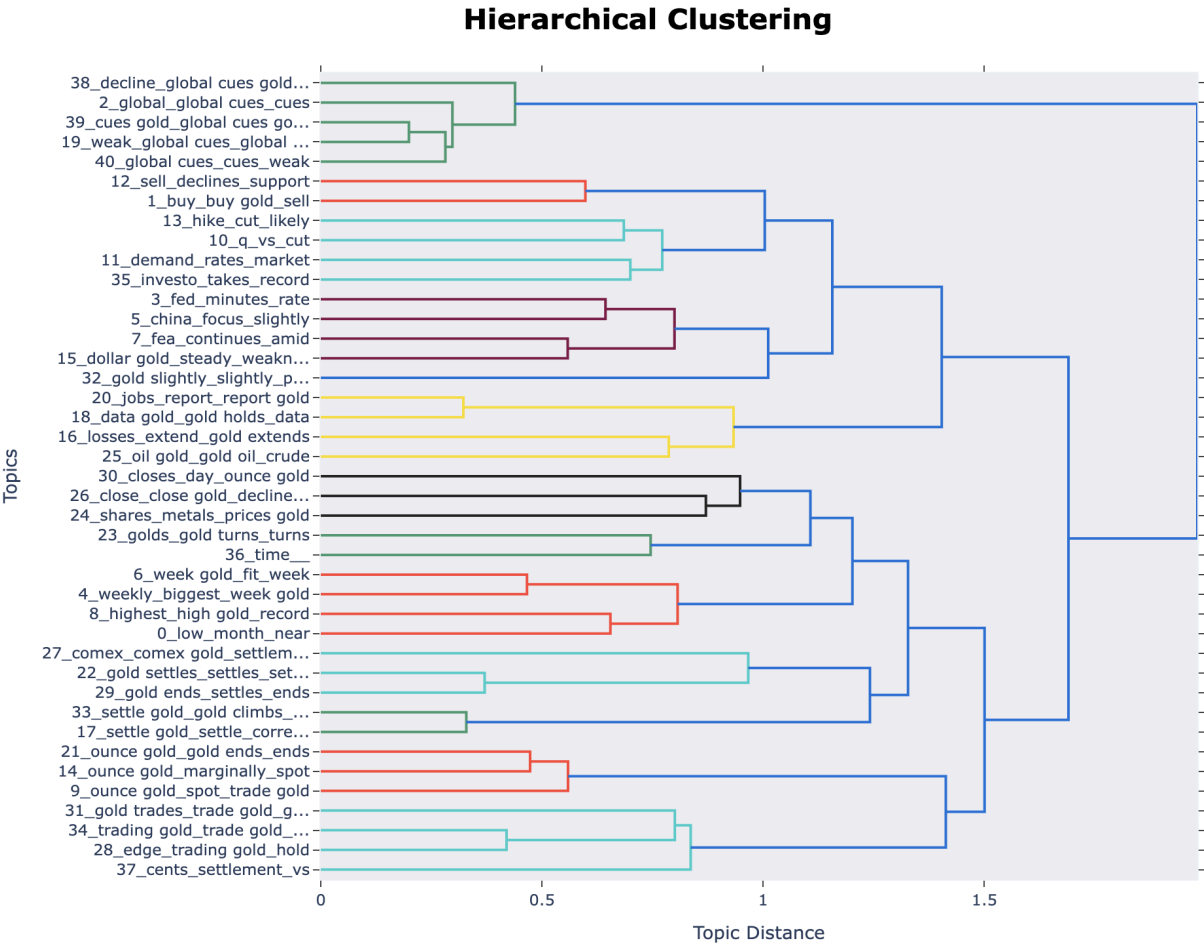


Figure 9: Visualizing Thematic Structure: Topic Distribution and Group Hierarchy

Upon observing overlapping semantics across topic clusters, we consolidated related topics into broader macro-themes. Using a cosine similarity matrix, we measured the pairwise relatedness of topic

embeddings. A high similarity score indicated semantic overlap, signaling which topics could be merged. We then applied hierarchical agglomerative clustering (HAC), a bottom-up approach that iteratively merged the most similar clusters until coherent macro-groups emerged. Together, these steps systematically resolved redundancy and revealed higher-level thematic structure. The optimal number of groups was dictated by the Davies-Bouldin score, which helped us balance intra-cluster cohesion and inter-cluster separation. Each headline's topic probability was then aggregated into macro group probabilities and converted to log-space to stabilize values for our downstream models (Appendix 11).

Given the scale of our dataset manually interpreting each cluster would have been infeasible. We employed Gemini-2.0-Flash, a lightweight large language model to automatically label each macro group for us. For every cluster we extracted 25 keywords and sampled representative headlines. Gemini returned concise, interpretable theme labels such as 'Comex Gold Closing Prices', 'Global Cues and Gold Stability', 'Rate Hikes/Cuts Price Impact' etc. This added a semantic layer for instant cluster interpretability while retaining full log-probability vectors per headline to preserve the multi-thematic nature of financial news (Appendix 12).

Having structured our dataset we now move from unsupervised exploration to predictive modeling. The thematic clusters we derived from financial headlines now serve as a high level proxies for market sentiment and macroeconomic context. We then aggregate the topic probabilities on a daily basis and create a frequency vector for each macro group that acts as our new model feature. This feature is then added to our existing baseline explanatory variable, i.e open price.

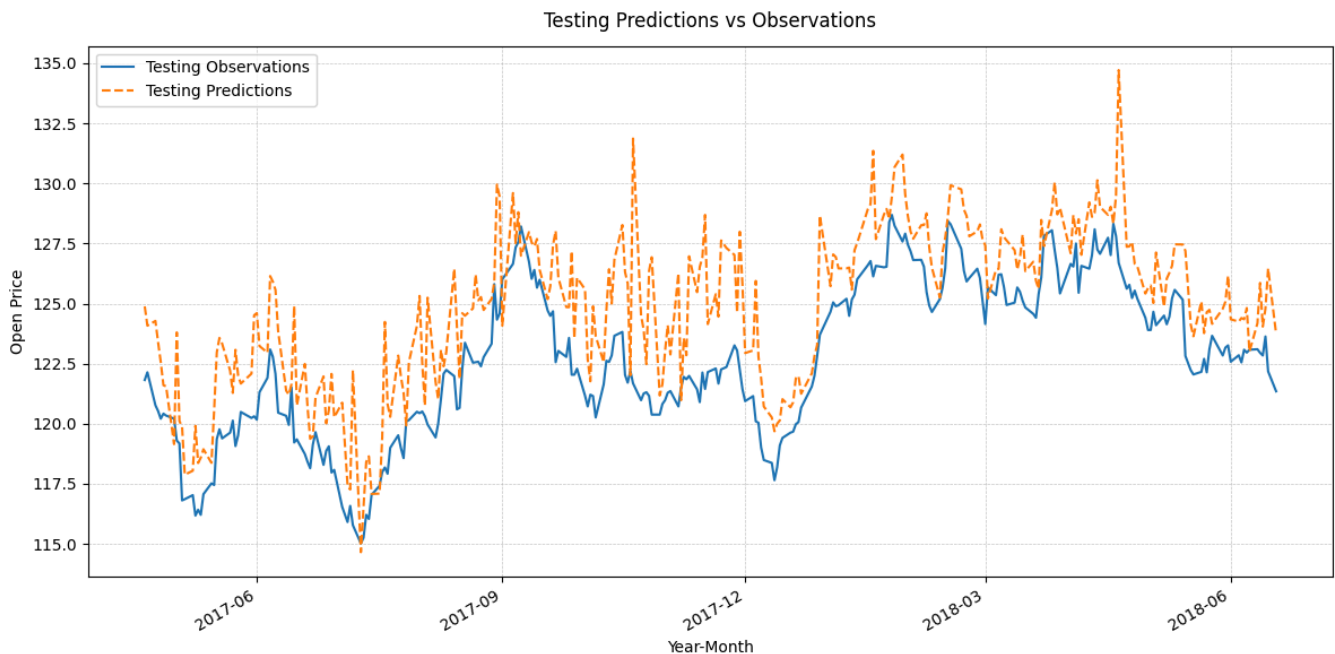


Figure 10: LSTM with Sentiment Performance: Validation vs Test set

Once prepared, we trained the LSTM model using the same architecture as previous experiments (Appendix 8), ensuring consistency across evaluations. The model's performance improved notably over the baseline, as detailed in Table 3.1 and Figure 10:

Table 3.1. Baseline Model vs Baseline with Cluster Group Metrics

| Model        | MAE    | MAE % | Directional Accuracy (%) |
|--------------|--------|-------|--------------------------|
| Baseline     | 0.5984 | 0.49% | 47.77%                   |
| w/Clustering | 2.2577 | 1.84% | 50.52%                   |

**Experiment 4:**

Having noticed that each of our additional features led to a better result we decided to build our final LSTM model that would incorporate a comprehensive set of features including historical open price data, sentiment scores, and topic frequency embedding derived from financial news. Once prepared, we trained the LSTM model using the same architecture as previous experiments (Appendix 8), ensuring consistency across evaluations.

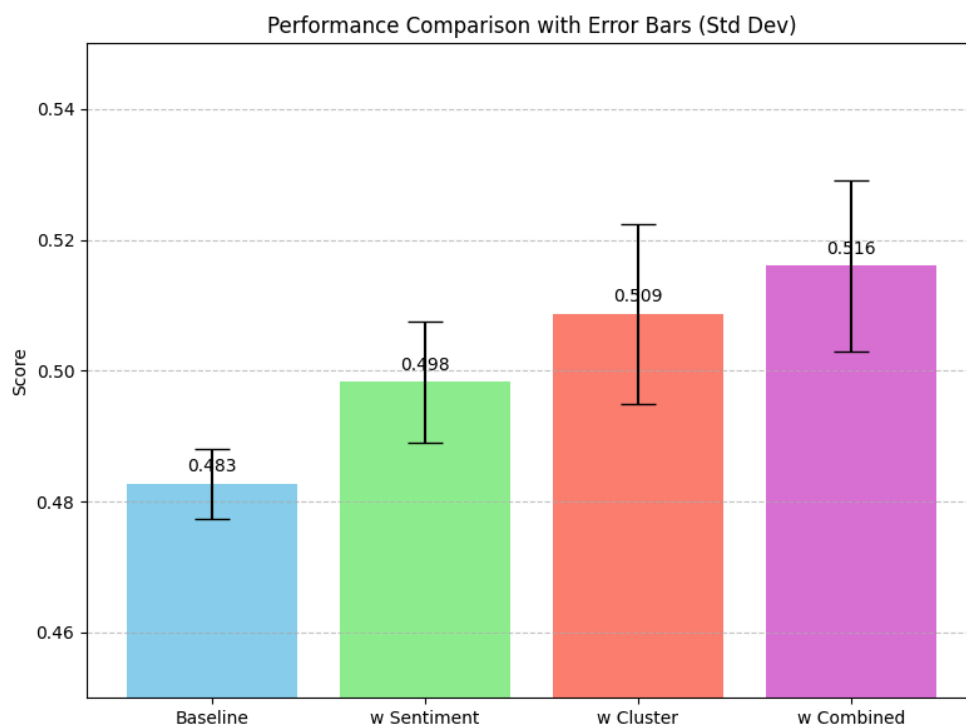


Figure 11: Model Performance Comparison

After running multiple iterations (i.e. ~10) of the models we developed, we noticed that not not only is the combined model better on average but when we account for uncertainty, it still consistently outperforms the other models as seen in Figure 11 (Appendix 13). This shows that it consistently outperforms the other models, making the case for the combined approach very strong.

In contrast, previous research on predicting daily directional movements in gold prices has primarily relied on semantic features, often augmented with traditional market indicators. These typically include Simple Moving Averages of gold, prices of related commodities (e.g., oil and silver), foreign exchange rates, and other financial metrics. While these additions have yielded modest improvements in predictive performance, our approach diverges in how clustering features are derived—utilizing semantic representations in a novel way. A summary of model types and their comparative performance is provided below for reference.

Table 4.1. Literature Review & Performance Comparison of Price Predictive Models

| Authors                                                         | Method                                            | Input variables     | Performance                                                                          |
|-----------------------------------------------------------------|---------------------------------------------------|---------------------|--------------------------------------------------------------------------------------|
| <a href="#">Kia, Haratizadeh, &amp; Shouraki, 2018</a>          | support vector machine + semi-supervised approach | previous gold price | Directional Accuracy = 56.8%                                                         |
| <a href="#">Parisi, A. A., Aparisi, F., &amp; Díaz, D. 2008</a> | Neural Network                                    | Gold Daily Prices   | Directional Accuracy = 60.68%                                                        |
| <a href="#">Livieris et al., 2020</a>                           | CNN + LSTM                                        | previous gold price | RMSE = 0.008,<br>MAE = 0.008,<br><br>Directional Accuracy= 0.516,<br><br>AUC = 0.519 |
| <a href="#">Weichen Gong 2023</a>                               | LSTM + Linear Regression                          | Gold daily prices   | LSTM Directional Accuracy = 50.67%<br><br>Linear Model Directional Accuracy = 53.02% |

### Experiment 5: Final Real Time analysis

Building on the above conclusions, we evaluated the model on a held-out test set, where it achieved a directional accuracy of 52.41%, indicating that it could capture overall trends and short-term movements better than random guessing by learning meaningful temporal patterns.

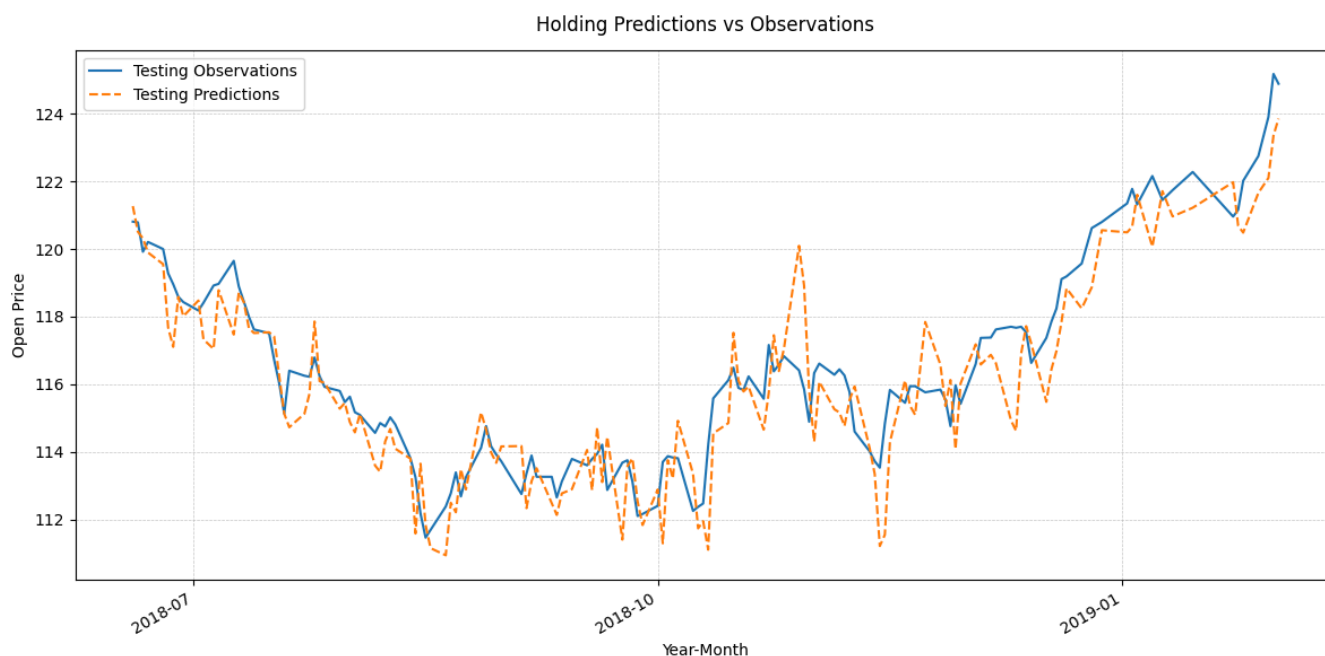


Figure 12: Hold-Out Test Performance

To improve performance, we retrained the model by merging the previous validation and test sets into the training data, while preserving a separate holdout set for final evaluation. This retraining resulted in a higher directional accuracy of 57.93%. The improvement can be attributed to two key factors: (1) the addition of more data, which allowed the LSTM architecture to better capture long-term dependencies and historical gold price behaviors, and (2) the temporal proximity of the holdout set to the training period, which leveraged the LSTM's strength in modeling sequential data where recent patterns influence future movements.

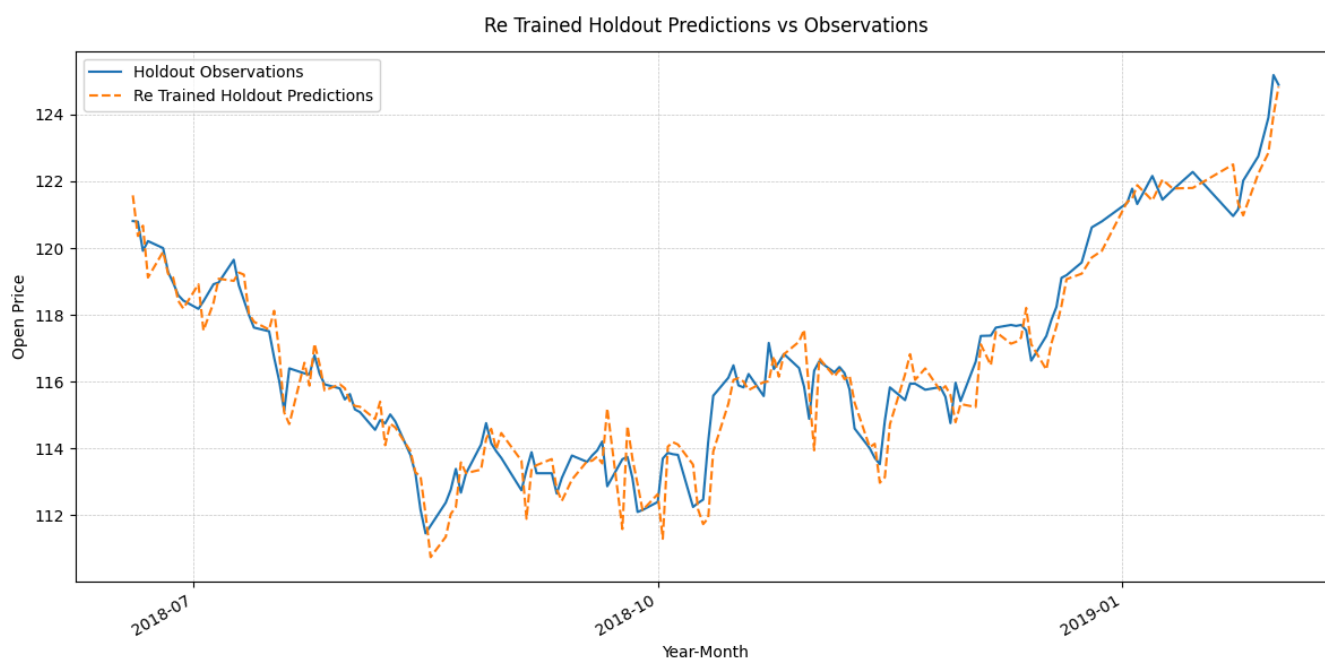


Figure 13: Hold-Out Train Performance

By generating a real time prediction using newsdata.io API to fetch recent gold related headlines along with yfinance to obtain corresponding gold prices, we were able to extend our model to predict gold prices for 2025. Our model predicted the gold price to be \$313.9 while the actual price was \$309.7, correctly predicting a price increase from the previous day (\$304.2).

## Results

The initial baseline LSTM model, which relied solely on historical gold prices, achieved modest results with directional accuracy below 50%, confirming the challenges of using price action alone. However, our step-by-step enhancement approach revealed that incorporating sentiment analysis from financial news headlines increased directional accuracy to 50.86%, while thematic clustering pushed performance to 50.52%. Most significantly, our combined model integrating all features achieved a directional accuracy of 52.58%, further improved to 57.93% after retraining with expanded data.

These findings suggest that market inefficiencies can indeed be captured through alternative data signals, particularly when sentiment and thematic context are considered alongside traditional indicators. Our real-time implementation, which correctly predicted a price increase with reasonable numerical accuracy despite being trained on data ending in 2020, demonstrates the practical applicability of our approach in dynamic market conditions.

## Ethical Considerations

Our sentiment-driven gold price prediction project faces ethical considerations including demographic and political biases in news platforms, exclusion of non-English signals in a global commodity market, and potential inherited biases in NLP models. While using public data, we'll address privacy through sentiment aggregation, platform terms compliance, and data minimization practices. Though our work may enhance market efficiency, widespread adoption could create feedback loops, amplify volatility, harm commodity-dependent economies, and disadvantage smaller participants. We'll mitigate these concerns through transparent communication of model limitations, explainable AI techniques, thorough documentation, reproducible code, clear uncertainty indicators, and contextual explanations of sentiment sources in our forecasts.

## Conclusions

This project examines the integration of sentiment from financial news into gold price prediction, demonstrating the capability of natural language processing to be used as a feature of traditional time-series models.

While our LSTM model demonstrates a promising directional accuracy of 57%, placing us on the profitable side, relying solely on the LSTM would be a risky venture. To fully capitalize on the model's edge and build a deployable, tradable system, it is essential to integrate additional components such as position sizing, the inclusion of other market indicator strategies, alternative datasets and risk

management frameworks. Without these elements, even a model with better accuracy could suffer from inconsistent returns and the compounding effects of losses, ultimately leading to poor financial outcomes.

Our research shows that even in largely efficient markets, qualitative signals—when properly quantified—can enhance predictive performance. This underscores the value of interdisciplinary methods that blend finance, machine learning, and natural language processing in today’s data-driven financial landscape.

## Team Roles

**Arko Bhattacharya:** Focused on Sentiment Clustering and Feature Engineering iterations of Models. Additionally, he structured our report and provided edits throughout the report to ensure consistency, clarity, and correctness.

**Zachary Fennie:** Focused on baseline model discovery, including implementation and performance analysis. Furthermore, he drafted and edited the paper including writing ethical considerations and conducting literature review.

**Vishesh Gupta:** Focused on Sentiment Clustering and Feature Engineering iterations of Models. Additionally, he structured our report and provided edits throughout the report to ensure consistency, clarity, and correctness.

**Danish Maknojia:** Conducted baseline model selection and evaluation, sentiment analysis and machine learning implementation, and documented the technical components of our models. Furthermore, drafted appropriate sections of the paper and literature review including the foundational background of the report.

**Christian Moreira:** Responsible for researching literature and working in the experiments that evaluate the intersection of sentiment metrics and market indexes - both against gold price variation. This assessment includes producing some of these metrics as well as optimizing them against the baseline model.

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## Appendix

1. Percentage change provides a normalized measure of price movements that's comparable across different time periods regardless of the absolute price level, making it more suitable for detecting patterns and measuring investment returns. This approach also helps address non-stationarity in the

time series [22]. Additionally, StandardScaler was implemented to standardize the percentage change to zero mean and unit variance, which mitigates the impact of outliers commonly found in volatile financial markets and improves the training stability of neural networks by ensuring all input features have similar magnitudes [23]. This approach further enhances the model's adaptability across different time periods, as standardized percentage changes remain comparable regardless of shifts in absolute price levels or market regimes.

2. For our LSTM model, as seen in Figure 14 below, the training loss curve demonstrates consistent improvement over 50 epochs, with training loss decreasing from approximately 1.08 to 0.87, while validation loss remains relatively stable between 0.70 and 0.78.

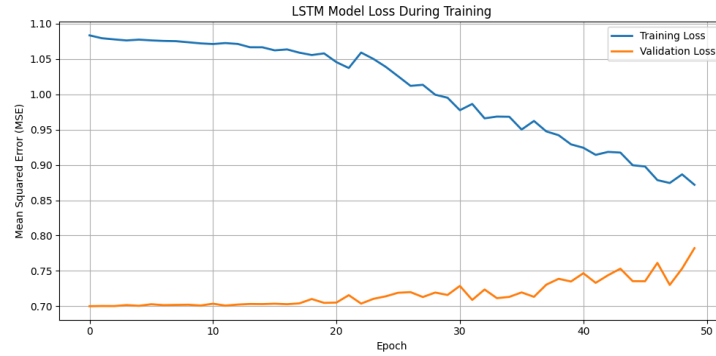


Figure 14: Long Short Term Memory Model Training Loss

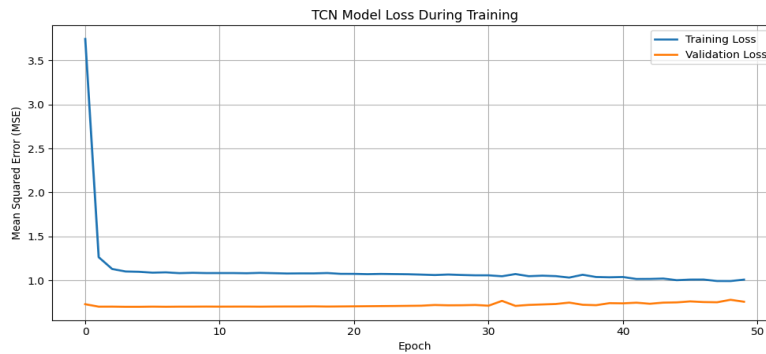


Figure 15: Temporal Convolutional Network Loss Training

Unlike the LSTM model, the TCN's training loss curve, in Figure 15, shows substantial learning during early epochs before stabilizing, with the training loss decreasing from approximately 4.0 to below 1.0, though the gap between training and validation loss suggests some degree of overfitting.

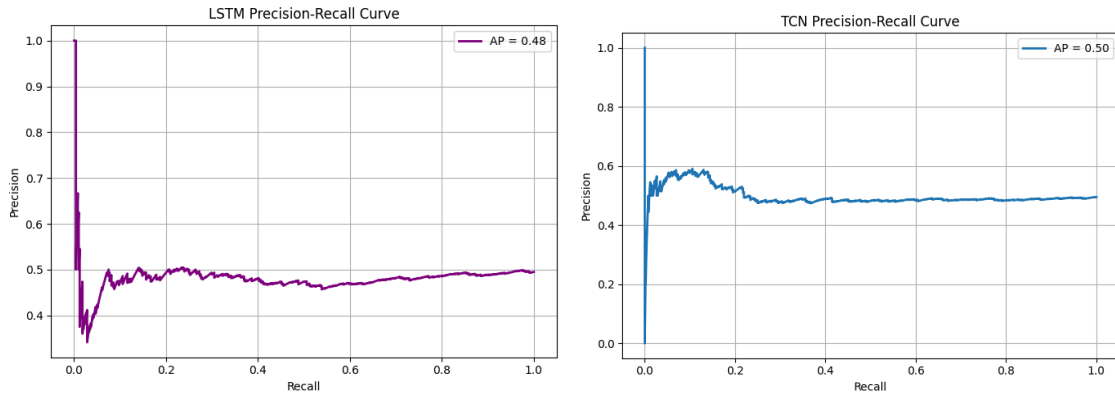


Figure 16: Long Short Term Memory & Temporal Convolutional Network Precision and Recall

3. Since percentage changes can introduce additional volatility and noise, modeling the “Open” price provided a more stable and structured signal, aligning more closely with the statistical assumptions of LSTM-based forecasting. Additionally, the opening price is a strong indicator of market sentiment at the beginning of each trading day, making it a meaningful and interpretable target for prediction.

4. We use the ticker "GLD" to obtain data for a gold-tracking Exchange Traded Fund (ETF). Data is pulled from January 1<sup>st</sup>, 2000, onward, using the yfinance library however we notice yfinance only has data from November 18<sup>th</sup>, 2004, onwards. Additionally, set `auto_adjust=False` to ensure that dividends and stock splits are not factored into price adjustments but are instead included as separate columns. The resulting dataset contains the following columns: Open, High, Low, Close, Adj Close, Volume, and, when applicable, Dividends and Stock Splits. We subset this data to include only the dates with headline coverage and focus solely on the Open price for our analysis.

5. We have a function that slides a window of size  $n+1$  over the input Dataframe. For each position it extracts the past  $n$  "Open" prices as a feature and the next "Open" price as the target. After declaring an  $n$  value and checking for enough rows it separates the window into inputs (first  $n$  prices) and targets (the last price). We flatten the inputs into a one-dimensional array and then store the corresponding date. Assuming  $n = 3$  after gathering all samples, it constructs a new Dataframe labeled "Target-3" to "Target-1" as seen in the table X below.

Table A1: Sample Target Pricing Window

| Target Date | Target-3 | Target-2 | Target-1 | Target |
|-------------|----------|----------|----------|--------|
| 2004-11-23  | 44.43    | 44.49    | 44.75    | 44.89  |
| 2004-11-24  | 44.49    | 44.75    | 44.89    | 44.93  |

This format is ideal for us to feed into our upcoming model to predict future prices based on past values.

6. We create a function that is used to generate the final input and output datasets for time series modeling. We provide it with a windowed Dataframe as seen above in Table X. The function extracts the first column as dates, which keeps track of when each prediction is meant to occur. Then it selects the middle columns—representing the past observed values used as features—and reshapes them into a 3D array with a shape that is expected for our LSTM model. Lastly, the final column is extracted as the target variable Y. We convert X and Y to float32 to make it compatible and efficient in training.
7. The first 75% of the data is used for training the model, allowing us to learn patterns and relationships from historical data. The next 10% is reserved for validation, which gives us space during model development to tune hyperparameters and prevent overfitting. The final 10% is kept for testing, which acts as a completely unseen dataset to objectively assess the model's performance in a real-world scenario. Lately 5% is kept as hold out data after our complete experiment has been run.
8. Our model is structured as a sequential neural network that has been optimized for time series forecasting. We first have an LSTM layer with the input shape as (x,y) that indicates that the training sample consists of x time steps (past prices) and y feature/s. Additionally the LSTM layer contains 64 memory units enabling it to learn complex patterns over time. We then include two densely connected layers (each with 32 neurons and ReLU activation) to our model. This helps us in learning nonlinear relationships and refining the representations extracted from the LSTM output. Lastly, we have a layer that is a single neuron with no activation that outputs the predicted "Open" price for the next time step. The model is then compiled using Mean Squared Error (MSE) as the loss function and the Adam optimizer with a learning rate of 0.001. Additionally, we track Mean Absolute Error (MAE) as a secondary performance metric.
9. We collect over 10,000 gold-related financial headlines that we have obtained via kaggle. Our preprocessing involves removing month names, numeric values, ticker symbols, and directional words (e.g., "rises," "falls"). This allows us to remove possible noise while preserving economic signals.
10. We further detail the configurations and components of BERTopic pipeline that we described above that was used to extract thematic structure from financial news headlines. The embedding model employed - all-mpnet-base-v2, is a sentence transformer that has been chosen for its strong performance in encoding short, noisy financial texts. We use a CountVectorizer to extract unigrams to trigrams with strict frequency thresholds (min\_df=10, max\_df=0.5, max\_features=5000) to reduce sparsity and highlight salient patterns. Dimensionality reduction was performed using UMAP which preserves both local and global structure in the high-dimensional embedding space. For clustering we used HDBSCAN that does not require a predefined number of clusters and is well-suited for discovering variable-sized clusters in noisy data. The model returned a soft assignment probability matrix over topics using HDBSCAN's all\_points\_membership\_vectors. This matrix, normalized row-wise, allowed us to understand the degree to which each headline belonged to different topics.
11. To address the semantic overlap among the topics we decided to introduce a macro level structure by grouping similar topics into broader thematic categories. We began by extracting topic embeddings generated by BERTopic, which are dense vector representations capturing semantic meaning. A cosine similarity matrix was computed between all topic embeddings to measure pairwise closeness. Using this matrix we performed hierarchical agglomerative clustering with Ward linkage, which minimizes the variance within merged clusters at each step. This helped us build a dendrogram that visualized how

topics were related hierarchically. We then evaluated the dendrogram cutoffs to determine the number of macro groups. For each potential threshold, we computed the Davies-Bouldin score. The threshold that minimized this score was selected, yielding an optimal macro-grouping structure. Once macro groups were defined, we aggregated topic-level soft assignments into macro-level probabilities. This was achieved by multiplying the topic probability matrix (output by BERTopic) with a one-hot encoding of the macro group mappings, effectively summing the probabilities of all component topics under each macro umbrella. Lastly, we applied a log transformation to the values that improved interpretability by compressing outlier values and better representing probabilistic uncertainty.

12. Our framework uses structurally derived Macro\_Group assignments for theme labeling and probability vector construction. Additionally, we compute a Dominant\_Macro\_Group for each headline based on the highest log-probability. These two features often diverge which highlights how structural and semantic affiliations can differ. Including both as features allows the model to learn from the topic hierarchy (i.e Macro\_Group) and from soft clustering signals (i.e Dominant\_Macro\_Group), improving generalization to ambiguous or noisy inputs.

13. Table 4.1. Baseline Model vs Baseline with Cluster Group Metrics

| Model        | MAE    | MAE % | Directional Accuracy (%) |
|--------------|--------|-------|--------------------------|
| Baseline     | 0.5984 | 0.49% | 47.77%                   |
| w/Sentiment  | 1.0833 | 0.88% | 50.86%                   |
| w/Clustering | 2.2577 | 1.84% | 50.52%                   |
| Combined     | 1.7277 | 1.41% | 52.58%                   |

14. These limitations are consistent with the Efficient Market Hypothesis (EMH), which suggests that historical price information alone offers limited predictive power, as asset prices already incorporate all publicly available information.

15. In finance the Efficient Market Hypothesis (EMH) is a foundational concept which states that assets prices fully reflect all available information at any given time therefore it is theoretically impossible to consistently achieve returns that outperform the overall market. Under this concept any attempt to beat the market based on existing information should ideally fail. Alpha, refers to the excess return of an investment relative to the return of a benchmark index or market. In the context of forecasting models, generating alpha means producing predictions that can be consistently turned into profitable trading strategies.

16. Averaging the logits rather than probabilities or class labels we are able to retain richer information about the model's confidence and uncertainty for each class. It allows you to preserve the strength of the sentiment signals across sentences before any distortions from nonlinear transformations like softmax. Additionally, averaging probabilities can introduce bias because probabilities are already forced to sum to 1 and are nonlinear — small differences in logits can become exaggerated or suppressed when passed through softmax.